



Assignment Overview: Data Integration

Estimated time: 4 minutes

Learning objectives

- Develop and execute data transformation and validation scripts using Linux shell to support data integration for fraud detection.
- Automate MySQL database backup operations using Bash scripting to ensure data reliability and recovery.

Introduction

In phase 1, you critically evaluated the existing data architectures of **FashionMart** and **TrendyThreads**. This evaluation uncovered key challenges such as data silos, inefficiencies, and integration issues. You reviewed data storage mechanisms and assessed data quality, integration points, security protocols, and governance practices to identify areas for improvement in the merged organization.

In phase 2, you used these insights to design a consolidated data architecture for **FutureMart**. This included developing data architecture diagrams, entity-relationship (ER) diagrams, database table structures, and initial schemas to address the previously identified gaps and performance bottlenecks.

In phase 3, you carried out the actual data migration between systems. This involved transferring data between RDBMS and NoSQL platforms, ensuring data integrity and consistency across staging environments to support a seamless transition to the new architecture.

In this module, you will work on the tasks of phase 4.

Phase 4: Integrate data systems for seamless data flow

In this phase, you will focus on integrating **FutureMart's** data systems to enable smooth and reliable data flow across platforms.

You will:

- Develop ETL/ELT scripts to move and transform data efficiently.
- Build robust data pipelines to connect various data sources and systems.
- Design a scalable data warehouse schema to support real-time analytics and reporting.
- Implement a specialized pipeline to support fraud detection mechanisms, ensuring timely insights and secure transactions.

Tasks:

Exercise 1: Create ETL script using Bash

In this exercise, you will create an ETL (Extract, Transform, Load) script using Bash to process CSV data.

- Extract data from two separate CSV files.
- Clean data by removing duplicate rows from each file.
- Merge the clean data into a single CSV file.

Exercise 2: Create data analytics script using Bash

In this exercise, you will build a Bash script to perform basic data analytics on product launch data.

- Extract data from specific columns
- Delete the header row from the extracted data
- Filter data and store it
- Count the number of products for **FutureMart**

Exercise 3: Optimize data analytics using Bash script (Optional)

In this exercise, you will optimize the data analytics script by combining multiple steps into a single command using Linux pipes.

Exercise 4: Backup MySQL database using Bash

In this exercise, you will write a Bash script to perform a backup of a MySQL database server.

Declare variables: Define the following in your script:

- DBUSERNAME
- DBPASSWORD
- HOST
- DB_NAME
- BACKUP_DIRECTORY

Write a script: Implement the logic to back up the database using the variables.

Deliverables

By the end of Module 4, you will have completed the following deliverables:

- ETL script created using Bash: **exercise_1_etl.sh**
- Data analytics script created using Bash: **exercise_2_data_analytics.sh**
- Bash script to back up the database on the MySQL server:
exercise_4_mysql_backup.sh